

# VIDHUN VIJAYAKUMAR SUJA

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## SUMMARY

MSc Software Engineering candidate and Backend Developer specializing in distributed systems, cloud-native microservices, and high-throughput API design. Shipped production systems on AWS with P95 latency under 40ms and consistent high availability under sustained concurrent load. Deep expertise in concurrency models, fault-tolerant architecture, thread pool tuning, caching strategies, and formal verification of concurrent programs. Completed dissertation on weak memory concurrency models and interactive execution visualisation; paper submitted to CONCUR 2026 (37th International Conference on Concurrency Theory, Liverpool).

## TECHNICAL SKILLS

**Languages:** Java, Python, TypeScript, JavaScript, SQL

**Backend and Systems:** Spring Boot, FastAPI, RESTful APIs, Microservices, Circuit Breakers (Resilience4j), Service Discovery (Eureka), Rate Limiting, Idempotency, Retry Logic, Thread Pool Tuning

**Infrastructure and Cloud:** AWS (ECS, RDS, S3), Docker, Kubernetes, GitHub Actions CI/CD, Redis, Render, Vercel

**Databases and Performance:** PostgreSQL, MySQL, Redis, Connection Pooling, Async I/O, k6 Load Testing

**Distributed Systems:** Weak Memory Models, Formal Verification, Concurrency, Race Condition Analysis, Graph Algorithms, Fault Tolerance, High Availability, Exponential Backoff

## EXPERIENCE

### Full-Stack Software Engineer Intern

*Kompetenzen Technologies, India | Dec 2023 - Aug 2024*

- Contributed to multiple production modules across a distributed Java/React platform, driving 30% reduction in P95 API latency and 15% uplift in user engagement through performance profiling, thread pool tuning, and executor optimization
- Designed high-throughput RESTful APIs serving 500+ concurrent sessions, increasing data processing throughput by 25% via connection-level optimizations and async processing patterns; load-tested with k6 to validate performance under sustained traffic
- Reduced concurrent user session failures by 95% through redesigning session management in Spring Boot, applying idempotent request handling and exponential backoff strategies, restoring 99%+ uptime and improving backend stability by 40%

## PROJECTS

### Software for Interactive Exploration of Concurrent Programs' Executions (MSc Dissertation) *Sep 2025 - Apr 2026*

**Stack:** Java, JavaFX, Graph Algorithms, Formal Methods, C11/C++11 Memory Models

*Supervised by Prof. Marko Doko, Heriot-Watt University*

- Built a Java 21/JavaFX desktop tool that lets developers step through concurrent program executions interactively — visualising memory operations as live graph nodes, with happens-before and synchronises-with edges updating in real time as each execution choice is made
- Closes a gap left by tools like herd7 and CDSChecker — rather than showing all possible executions at once, the tool guides step-by-step construction of each execution, making it significantly easier to reason about why a specific outcome is allowed or forbidden on ARM and POWER hardware
- Conducted a structured user study (n=9) measuring time-to-understanding for advanced concurrency concepts; paper submitted to CONCUR 2026 (37th International Conference on Concurrency Theory)

### AI Resume Analysis Platform (Production)

*Jan 2026 - Feb 2026*

**Stack:** Python, FastAPI, React, TypeScript, OpenAI GPT-3.5, PostgreSQL, Redis, Docker, AWS, Render, Vercel

**Live:** [ai-resume-analyzer-eight-xi.vercel.app](https://ai-resume-analyzer-eight-xi.vercel.app) **GitHub:** [github.com/vsvidhun06-blip/ai-resume-analyzer](https://github.com/vsvidhun06-blip/ai-resume-analyzer)

- Architected and shipped production AI platform: async FastAPI backend with PostgreSQL connection pooling, JWT auth, Redis caching reducing DB load by 35%, and rate limiting middleware to block abusive traffic patterns
- Load-tested with k6 to validate P95 latency under 40ms and P99 under 80ms under sustained concurrent load; deployed via zero-downtime CI/CD (GitHub Actions + Render + Vercel) with idempotent retry logic and exponential backoff on all OpenAI API calls

### Cloud-Native E-Commerce Microservices Platform

*Dec 2025 - Feb 2026*

**Stack:** Java, Spring Boot, Kafka, Docker Compose, PostgreSQL, Redis, Eureka, Resilience4j, GitHub Actions

**GitHub:** [github.com/vsvidhun06-blip/ecommerce-microservices](https://github.com/vsvidhun06-blip/ecommerce-microservices)

- Designed distributed e-commerce backend across independently deployable microservices orchestrated with Docker Compose, with tuned thread pools per service; implemented Eureka service discovery, Resilience4j circuit breakers, and Redis caching layer for consistent high availability under load

- Reduced deployment cycle time by 60% via automated CI/CD with blue-green deployments; built collaborative filtering Recommendation Engine targeting 15% improvement in conversion metrics, load-tested end-to-end with k6

## ACHIEVEMENTS

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- Competitive Programming: 153 LeetCode problems solved (33 Easy, 93 Medium, 27 Hard) across arrays, DP, backtracking, divide & conquer, graphs, and binary search — [leetcode.com/u/VidhuViny](https://leetcode.com/u/VidhuViny)

## EDUCATION

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**MSc Software Engineering — Heriot-Watt University, Edinburgh, UK**

*Expected May 2026*

*Distributed Systems, Formal Verification, Advanced Concurrency, System Architecture*

**BTech Computer Science and Engineering — College of Engineering Muttathara, Kerala, India** *July 2023*

## REFERENCES

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### **Prof. Marko Doko**

Dissertation Supervisor | Associate Professor, School of Mathematical and Computer Sciences

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